

ID:

Set A

Name:

Sec: 11 (8am)

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. $[1 \times 3 = 3]$

a) $\Sigma = \{a, b, c\}$, if there are 6 states, there must be 16 outgoing transitions in a DFA.

Ans: False. For dfa each state MUST have 3 outgoing transitions. So there is $(3 \times 6 = 18)$ total transition.

b) A regular language will always string of fixed length.

Ans: True

c) Epsilon transition can happen in both DFA and NFA.

Ans: False. DFA can never have epsilon transition.

2] Suppose there are two DFAs, DFA_1 and DFA_2 . DFA_1 has a total of 7 states with 3 accepting states and rest are rejecting states. And DFA_2 has total 5 states with 3 accepting states and rest are rejecting states. We get new DFA from the cross product of DFA_1 and DFA_2 . Find: $[1+2=3]$

a) The total number of states in the new DFA.

b) The total number of accepting states in the new DFA if we do $DFA_1 \cup DFA_2$.

a) Total states = $(7 \times 5) = 35$

b) Accepting states = $35 - (2 \times 4) = 35 - 8 = 27$ states

3] Write the Regex for the following language. $L = \{w \in \{a, b\}^* \mid w \text{ starts and ends with different symbol}\}$

[Marks = 2]

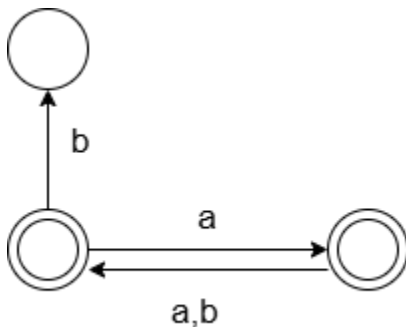
$a(a|b)^*b \mid b(a|b)^*a$

4] Draw the DFA for :- $L = \{w \in \{a, b\}^* \mid \text{every odd position in } w \text{ has } a\}$ [Marks - 4]

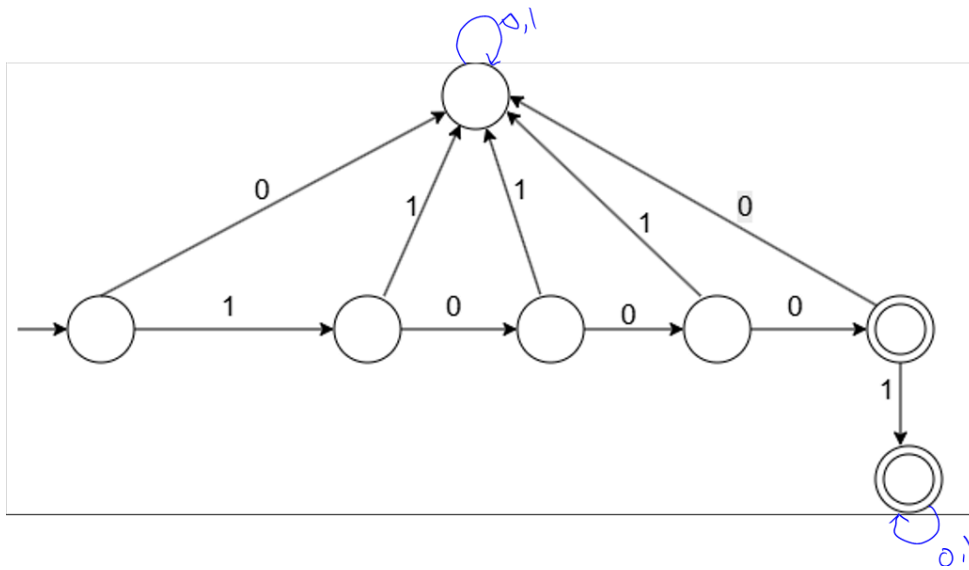
5] Draw the DFA for :- $L = \{w \in \{0, 1\}^* \mid \text{always starts with } 1, \text{ followed by at most three '0's}\}$ [Marks-4]

6] Draw the DFA for :- $L = \{w \in \{0, 1\}^* \mid \text{Does not start with '110'}\}$ [Marks - 4]

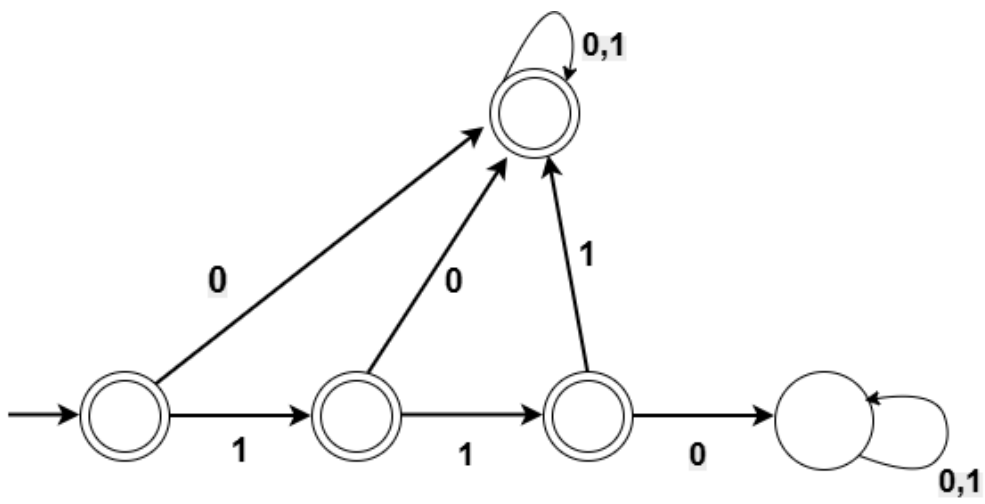
Ans-4).



Ans-5).



Ans-6).



ID:

Set B

Name:

Sec: 11 (8am)

1. Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1 x 3 = 3]

a) $\Sigma = \{a, b, c\}$, if there are 6 states, there must be 18 outgoing transitions in an NFA.

Ans: False. Nfa does not require outgoing transitions for each states, unlike DFA.

b) A regular language will always contain fixed number of strings.

False. Regular language can be infinite where there can be infinite number of strings.

c) The union operation of two regular languages will always create a regular language

True

2] Suppose there are two DFAs, DFA_1 and DFA_2 . DFA_1 has a total of 6 states with 2 accepting states and rest are rejecting states. And DFA_2 has total 5 states with 2 accepting states and rest are rejecting states. We get new DFA from the cross product of DFA_1 and DFA_2 . Find: [1+2 = 3]

a) The total number of states in the new DFA .

b) The total number of accepting states in the new DFA if we do $DFA_1 \cup DFA_2$.

a) Total states = $(6 * 5) = 30$

b) Accepting states = $30 - (4 * 3) = 30 - 12 = 18$ states

3] Write the Regex for the following language. $L = \{w \in \{0, 1\}^* \mid w \text{ starts and ends with same symbol}\}$

[Marks = 2]

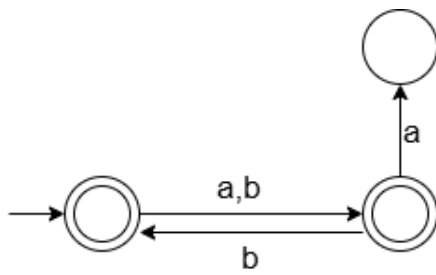
$1(0|1)^*1 \mid 0(0|1)^*0 \mid 0 \mid 1$

4] Draw the DFA for :- $L = \{w \in \{a, b\}^* \mid \text{every even position in } w \text{ has } b\}$ [Marks - 4]

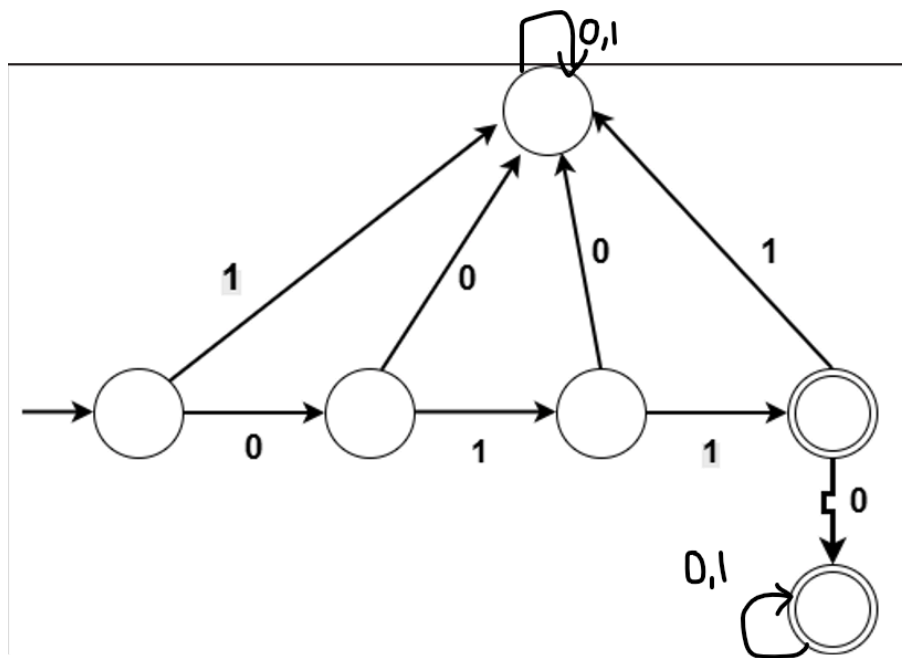
5] Draw the DFA for :- $L = \{w \in \{0, 1\}^* \mid \text{always starts with } 0, \text{ followed by exactly two '1's}\}$ [Marks-4]

6] Draw the DFA for :- $L = \{w \in \{a, b\}^* \mid \text{Does not start with 'abb'}\}$ [Marks - 4]

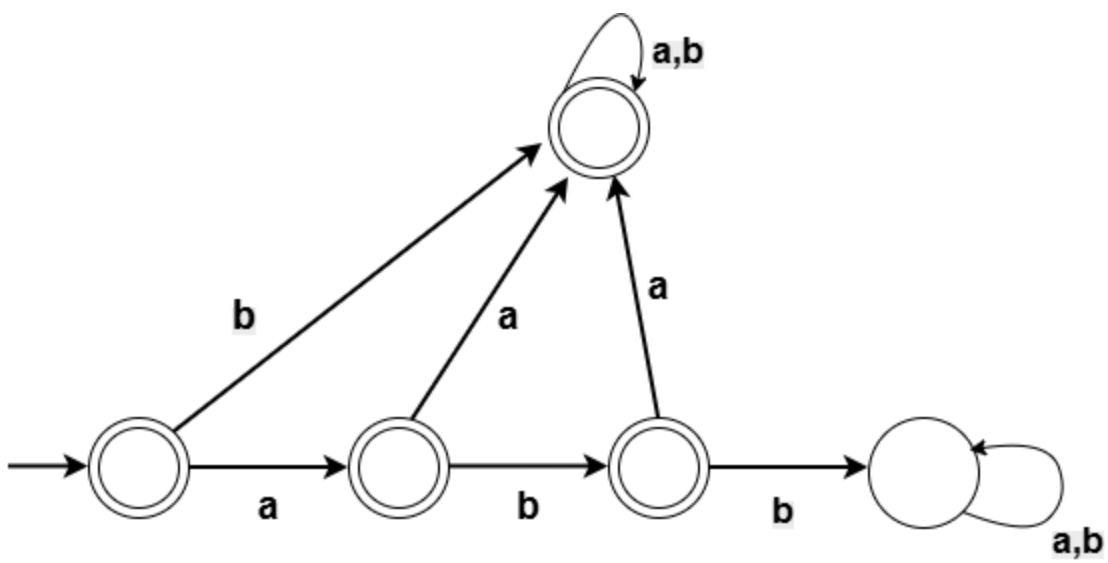
Ans-4).



Ans-5).



Ans-6).



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Sec:

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1 x 2 = 2]

a) $\Sigma = \{a, b, c\}$, if there are 6 states, there must be 16 outgoing transitions in a DFA.

Ans: False. For dfa each state MUST have 3 outgoing transitions. So there is $(3 * 6 = 18)$ total transition.

b) A regular language will always string of fixed length.

Ans: True

2] Suppose there are two DFAs, DFA_1 and DFA_2 . DFA_1 has a total of 7 states with 3 accepting states and rest are rejecting states. And DFA_2 has total 5 states with 3 accepting states and rest are rejecting states. We get new DFA from the cross product of DFA_1 and DFA_2 . Find: [1 + 2 = 3]

a) The total number of states in the new DFA .

b) The total number of accepting states in the new DFA if we do $DFA_1 \cap DFA_2$.

a) Total states = $(7 * 5) = 35$

b) Accepting states = $3 * 3 = 9$ states

3] Write the Regex for the following language. $L_2 = \{w \in \{0, 1\}^* \mid w \text{ contains odd number of '0's or even number of '1's}\}$ [Marks = 2]

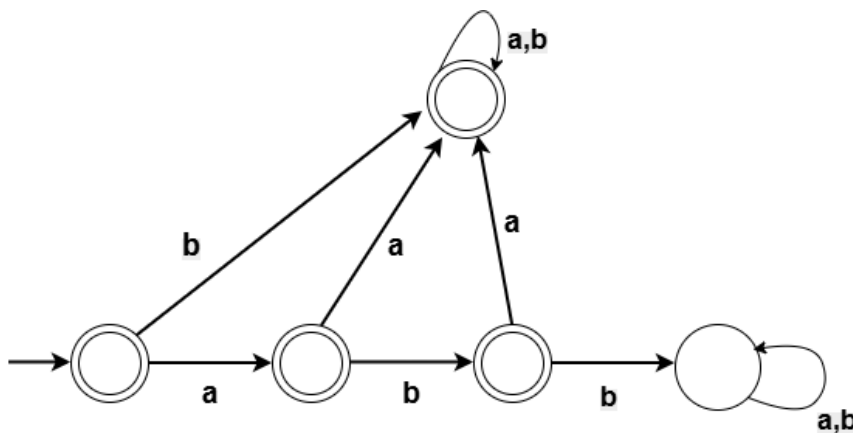
Ans: $(1^*01^*01^*)0 \mid (0^*10^*10^*)$

4] Draw the DFA for :- $L = \{w \in \{a, b\}^* \mid \text{Does not contain the substring with 'abb'}\}$ [Marks-4]

5] Draw the DFA for :- $L = \{w \in \{0, 1\}^* \mid \text{Every } (5k + 2)\text{th position must have '1', where } k \geq 0\}$ [Marks-5]

6] Draw the NFA for the language, L_3 , where $L_3 = L_2 \cdot L_1$. Here, $L_1 = \{\text{starts with 10}\}$ & $L_2 = \{\text{length is } 2k+1, \text{ where } k \geq 0\}$. Given, $\Sigma = \{0, 1\}$. [Marks-4]

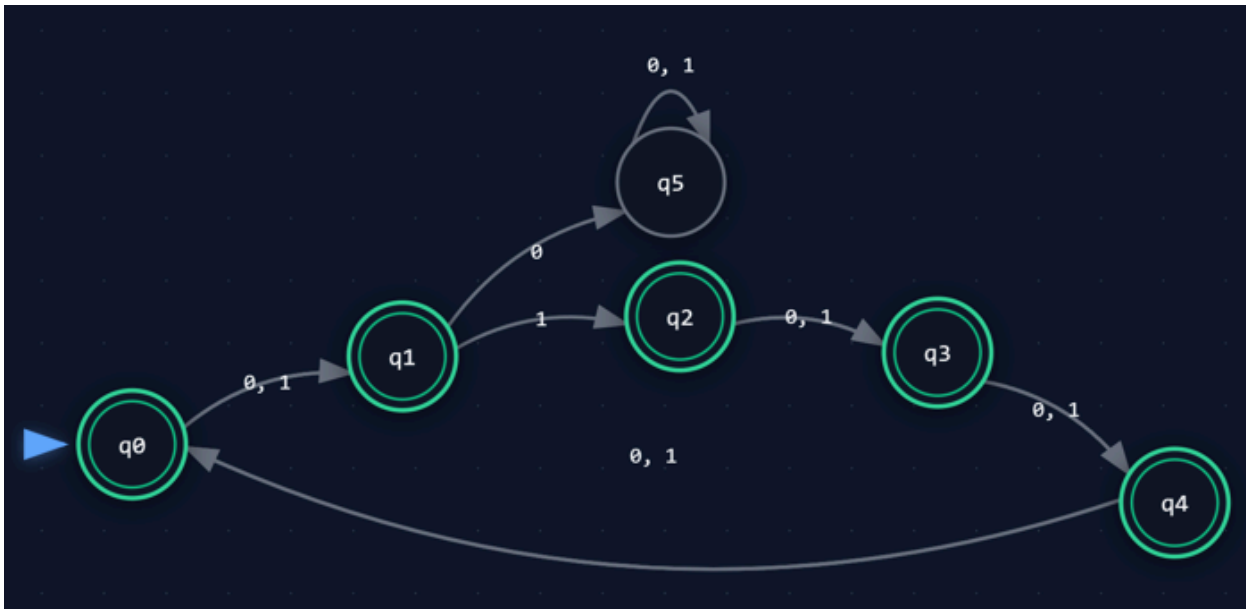
Ans-4).



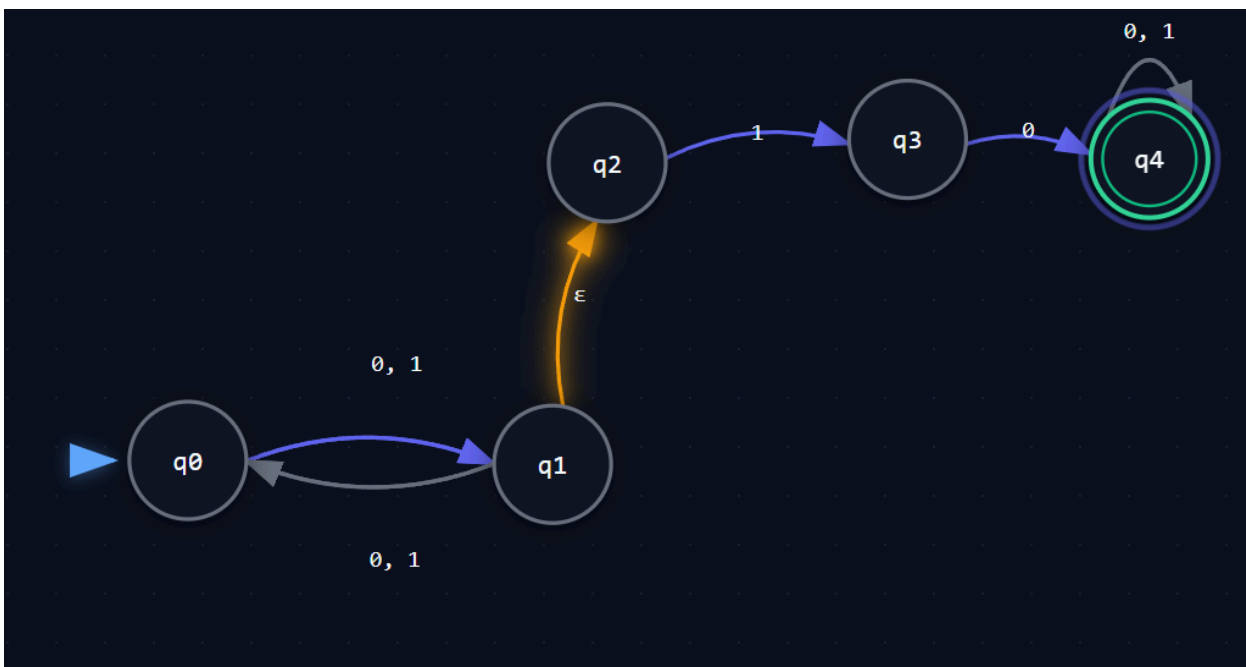
ID:
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Sec:

Ans-5).



Ans-6).



ID:

Name:

Sec:

1. Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1 x 2 = 2]

a) The union operation of two regular languages will always create a regular language

True

b) $\Sigma = \{a, b, c\}$, if there are 6 states, there must be 18 outgoing transitions in an NFA.

Ans: False. Nfa does not require outgoing transitions for each states, unlike DFA.

2] Suppose there are two DFAs, DFA_1 and DFA_2 . DFA_1 has a total of 6 states with 2 accepting states and rest are rejecting states. And DFA_2 has total 5 states with 2 accepting states and rest are rejecting states. We get new DFA from the cross product of DFA_1 and DFA_2 . Find: [1+2 = 3]

a) The total number of states in the new DFA .

b) The total number of accepting states in the new DFA if we do $DFA_1 \cap DFA_2$.

a) Total states = $(6 * 5) = 30$

b) Accepting states = $2 * 2 = 4$ states

3] Write the Regex for the following language. $L_2 = \{w \in \{0, 1\}^* \mid w \text{ contains odd number of '0's' or even number of '1's'}\}$ [Marks = 2]

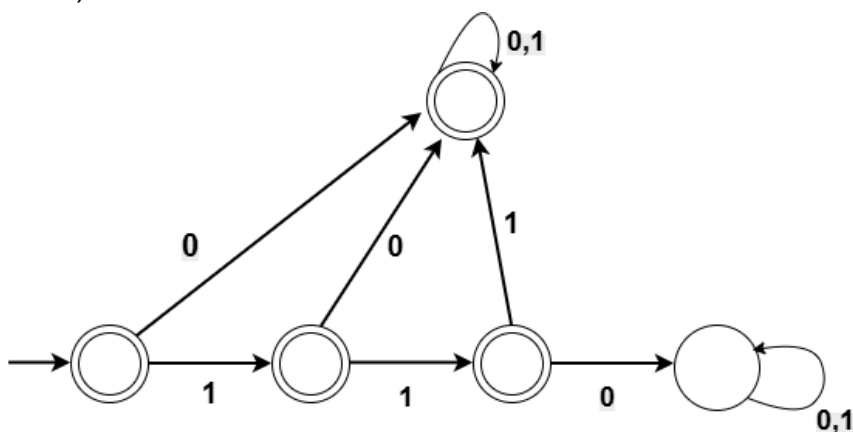
Ans: $(1^*01^*01^*)0 \mid (0^*10^*10^*)$

4] Draw the DFA for :- $L = \{w \in \{0, 1\}^* \mid \text{Does not contain the substring with '110'}\}$ [Marks-4]

5] Draw the DFA for :- $L = \{w \in \{a, b\}^* \mid \text{Every } (5k + 3)\text{th position must have 'a', where } k \geq 0\}$ [Marks-5]

6] Draw the NFA for the language, L_3 , where $L_3 = L_2 \cdot L_1$. Here, $L_1 = \{\text{starts with cc}\}$ & $L_2 = \{\text{length is } 2k+1, \text{ where } k \geq 0\}$. Given, $\Sigma = \{c, d\}$. [Marks-4]

Ans-4).

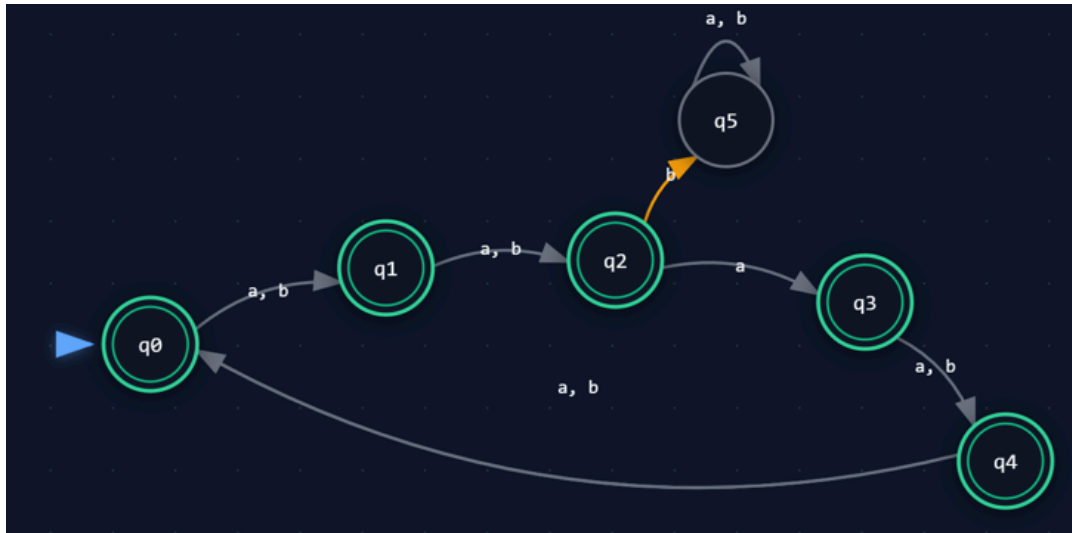


Ans-5).

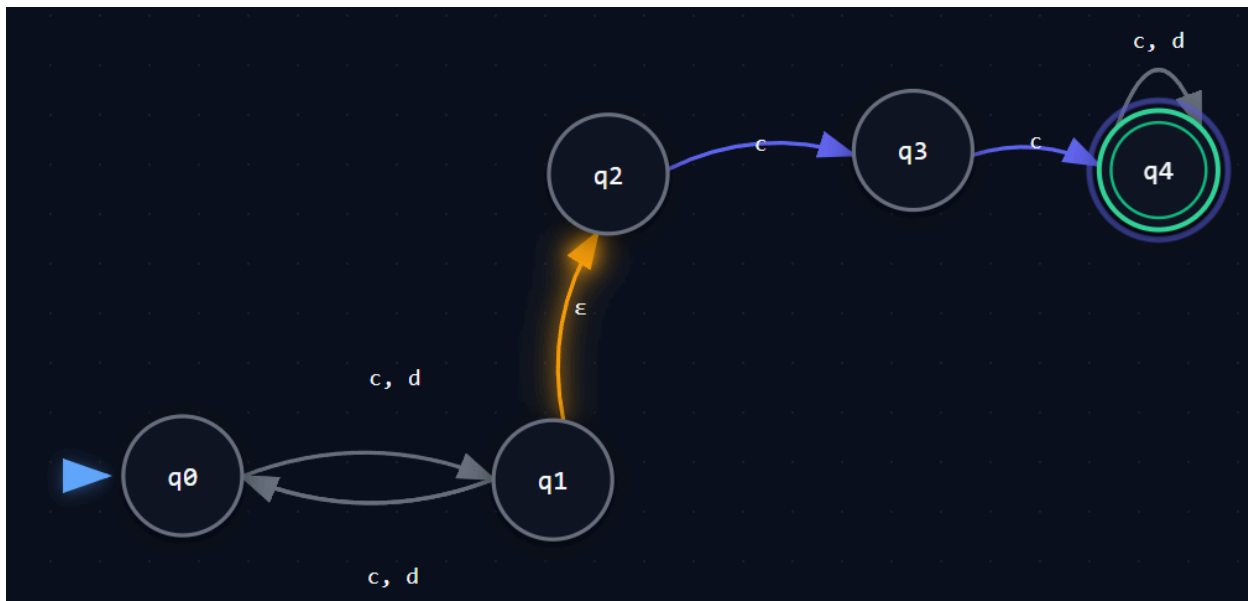
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Ans-6).



Sec11

ID:

Set A

Name:

Sec:

Qs 1) Solve the following problems: $[6 * 2 = 12]$

$L_1 = \{ w \in \{0, 1\}^* \mid \text{last two symbols of } w \text{ are different.} \}$

$L_2 = \{ w \in \{0, 1\}^* \mid w = 0^m 1^n \text{ where } m \geq 1 \text{ and } n \geq 2 \}$

$L_3 = \{ w \in \{0, 1\}^* \mid '1' \text{ is never followed by another '1'} \}$

$L_4 = \{ w \in \{0, 1\}^* \mid \text{if it starts with 0, it contains exactly three '1's'} \}$

$L_5 = \{ w \in \{0, 1\}^* \mid |w| = 4n + 2 \}$

$L_6 = \{ w \in \{0, 1\}^* \mid \text{even number of '0's'} \}$

$L_7 = \{ w \in \{0, 1\}^* \mid \text{exactly three '1's'} \}$

a) Regular expression for $L_1 = (0|1)^* 10 \mid (0|1)^* 01$

b) Regular expression for $L_2 = 00^*111^*$

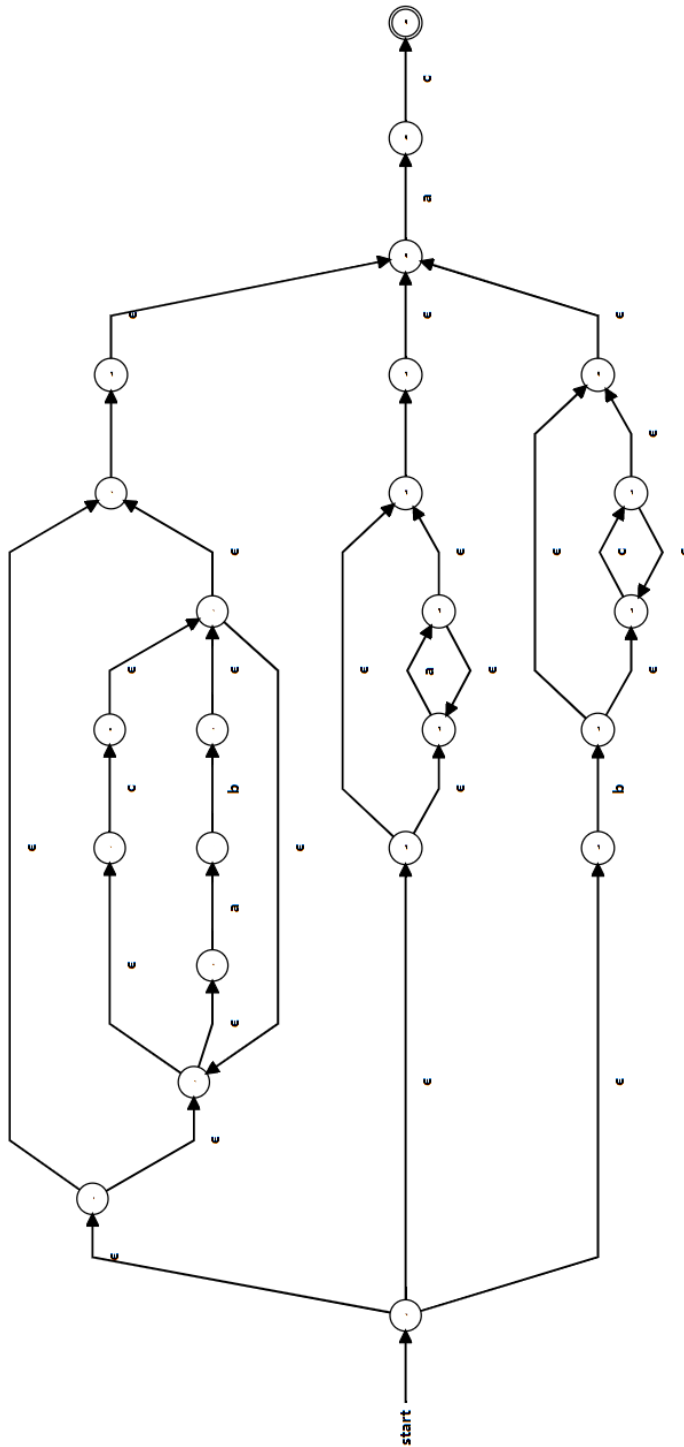
c) Regular expression for $L_3 = (10|0)^* (1 \mid \epsilon)$

d) Regular expression for $L_4 = 01110^* \mid 1(0|1)^*$

e) Regular expression for $L_5 = ((0|1)(0|1)(0|1)(0|1))^* (0|1)(0|1)$

f) Regular expression for $L_6 \cup L_7$ ($\cup$ denotes or). $= (1^*01^*01^*)^* \mid 0^*10^*10^*10^*$

Qs 2) Convert the following Regex to NFA over $\Sigma = \{a, b, c\}$. [8]

$$(((ab) + c)^* | (a^* + bc^*)ac$$


ID:

Set B

Name:

Sec:

Qs 1) Solve the following problems: $[6 * 2 = 12]$

$L_1 = \{ w \in \{0, 1\}^* \mid \text{last two symbols of } w \text{ are same.} \}$

$L_2 = \{ w \in \{0, 1\}^* \mid w = 0^m 1^n \text{ where } m \geq 2 \text{ and } n \geq 1 \}$

$L_3 = \{ w \in \{a, b\}^* \mid \text{'b' is never followed by another 'b'} \}$

$L_4 = \{ w \in \{0, 1\}^* \mid \text{if it starts with 1, it contains atleast three '0's'} \}$

$L_5 = \{ w \in \{0, 1\}^* \mid |w| = 4n + 3 \}$

$L_6 = \{ w \in \{0, 1\}^* \mid \text{odd number of '0's'} \}$

$L_7 = \{ w \in \{0, 1\}^* \mid \text{exactly two '1's'} \}$

a) Regular expression for $L_1 = (0|1)^* 11 \mid (0|1)^* 00$

b) Regular expression for $L_2 = 000^* 11^*$

c) Regular expression for $L_3 = (ba|a)^* (b|e)$

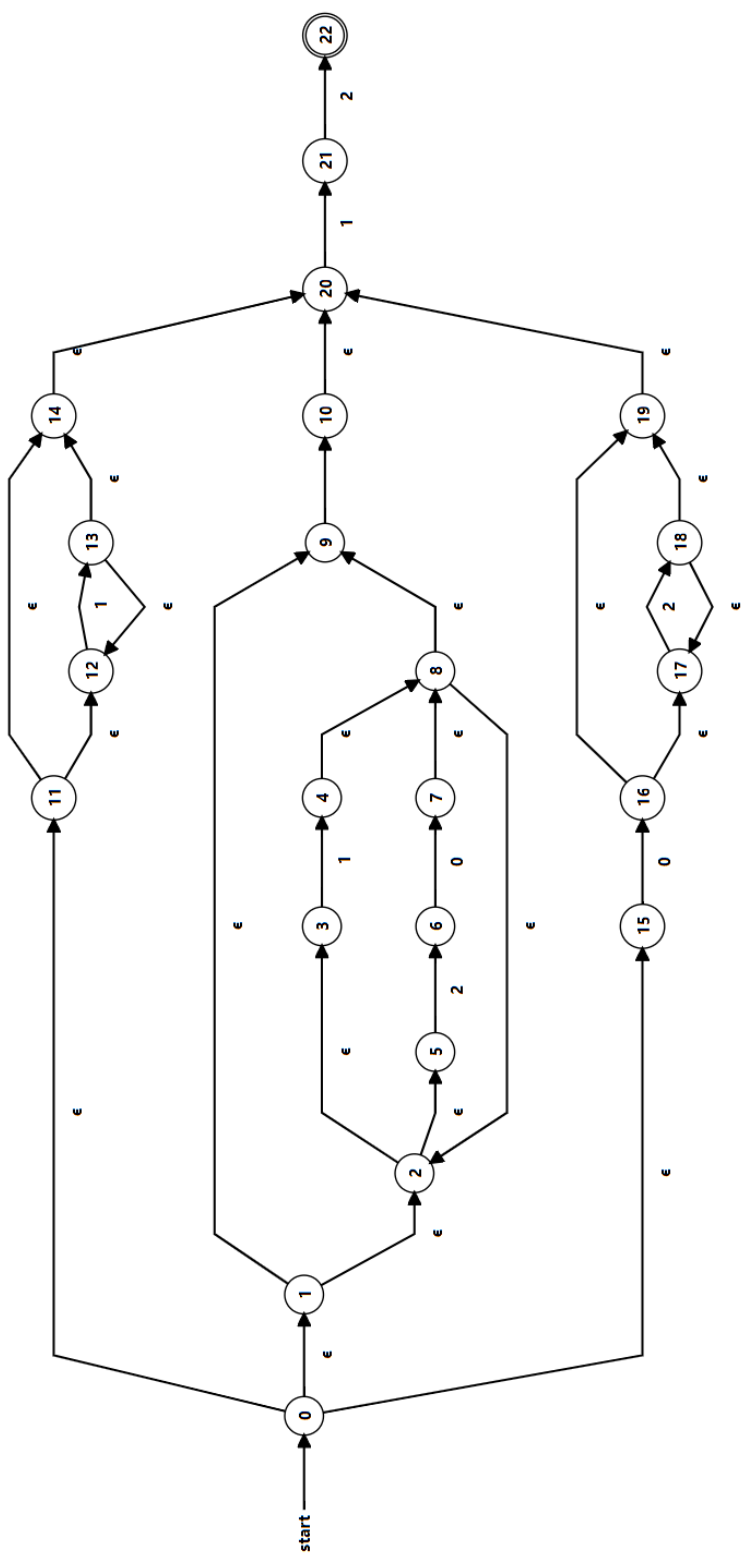
d) Regular expression for $L_4 = 10000^*(0|1)^* \mid 0(0|1)^*$

e) Regular expression for $L_5 = (\Sigma\Sigma\Sigma\Sigma)^* \Sigma\Sigma$

f) Regular expression for $L_6 \cup L_7$ ($\cup$ denotes or). $= (1^* 0 1^* 0 1^*)^* \mid (0^* 1 0^* 1 0^*)^*$

Qs 2) Convert the following Regex to NFA over $\Sigma = \{0,1,2\}$. [8]

$((1 + (20))^* |(1^* + 02^*)12$



Sec12

ID:

Set A

Name:

Sec:

Qs 1) Solve the following problems: $[5 * 2 = 10]$

$L_1 = \{ w \in \{0, 1\}^* \mid \text{if it starts with 1, the last two symbols of } w \text{ are different.} \}$

$L_2 = \{ w \in \{0, 1\}^* \mid \text{does not contains substring '11'} \}$

$L_3 = \{ w \in \{0, 1\}^* \mid \text{contains at least one 1 and odd 0's after the last 1} \}$

$L_4 = \{ w \in \{0, 1\}^* \mid |w| \% 3 \neq 1 \}$

$L_5 = \{ w \in \{0, 1, 2\}^* \mid w = 0^m 1^n 2^k \text{ where } m, k \geq 2 \text{ and } n \geq 1 \}$

a) Regular expression for $L_1 = 1(0|1)^*10 \mid 1(0|1)^*10 \mid 0(0|1)^*$

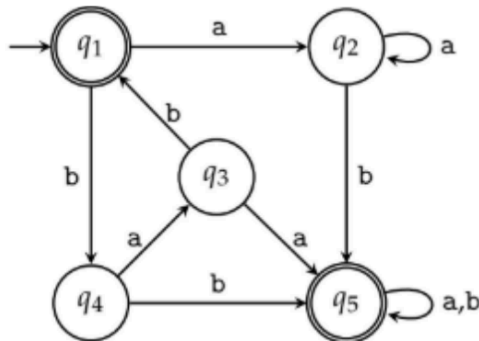
b) Regular expression for $L_2 = (10|01)^* (1|0)$

c) Regular expression for $L_3 = (0|1)^*1(00)^*0$

d) Regular expression for $L_6 = L_4'$ (denotes inverse) $= (\Sigma\Sigma\Sigma)^*\Sigma$

e) Regular expression for $L_6 \cdot L_5$ (. denotes concatenation) $= ((\Sigma\Sigma\Sigma)^* \mid (\Sigma\Sigma\Sigma)^*\Sigma\Sigma).000^*11^*222^*$

Qs2) Eliminate State q_4 [Marks = 2]



New state(S) having epsilon transition to q_1 . S is the new starting state.

A new state (F) is created. q_1 and q_5 has outgoing transition to F. F is the ONLY accepting state.

If we eliminate q_4 , let's consider

Incoming transition from $\rightarrow q_1$

Outgoing transition to $\rightarrow q_3, q_5$

So if q_4 is removed we have, q_1 to q_3 & q_1 to q_5 .

q_1 to $q_3 = ba$

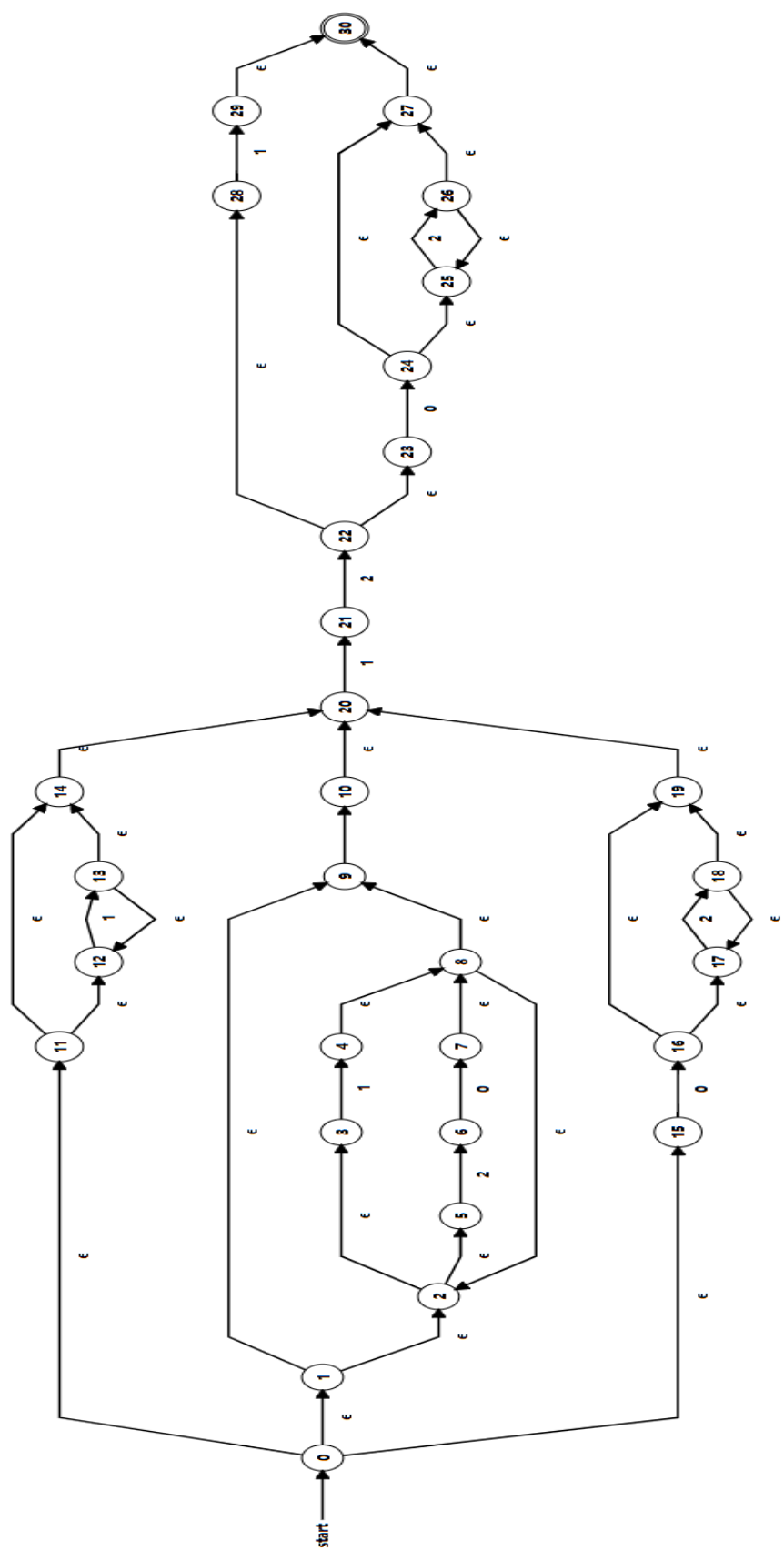
q_1 to $q_5 = bb$

Qs 3) Convert the following Regex to NFA over $\Sigma = \{0,1,2\}$. [8]

$((1 + (20))^* | (1^* + 02^*)12(3.2^* \cup 1)$

Correction

$((1 + (20))^* | (1^* + 02^*)12(\textcolor{red}{0}.2^* \cup 1)$



ID:

Set B

Name:

Sec:

Qs 1) Solve the following problems: $[5 * 2 = 10]$

$L_1 = \{ w \in \{a, b\}^* \mid \text{if it starts with a, last two symbols of } w \text{ are different.} \}$

$L_2 = \{ w \in \{0, 1\}^* \mid \text{contains at least one 0 and odd 1's after the last 0} \}$

$L_3 = \{ w \in \{a, b\}^* \mid \text{does not contain the substring 'aa'} \}$

$L_4 = \{ w \in \{0, 1\}^* \mid |w| \% 3 \neq 2 \}$

$L_5 = \{ w \in \{0, 1, 2\}^* \mid w = 0^m 1^n 2^k \text{ where } m, k \geq 1 \text{ and } n \geq 3 \}$

a) Regular expression for $L_1 = 0(0|1)^*10 \mid 0(0|1)^*10 \mid 1(0|1)^*$

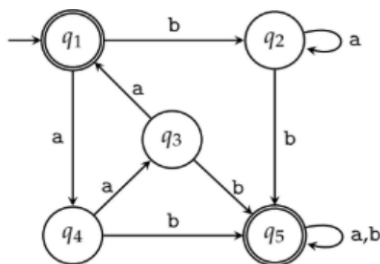
b) Regular expression for $L_2 = (0|1)^*0(11)^*1$

c) Regular expression for $L_3 = (ab|b)^*(a|e)$

d) Regular expression for $L_6 = L_4'$ (denotes inverse) $= (\Sigma\Sigma\Sigma)^* \mid (\Sigma\Sigma\Sigma)^*\Sigma$

e) Regular expression for $L_6 \cdot L_5$ (. denotes concatenation) $= ((\Sigma\Sigma\Sigma)^* \mid (\Sigma\Sigma\Sigma)^*\Sigma).00^*1111^*22^*$

Qs2) Eliminate State q_4 [Marks = 2]



New state(S) having epsilon transition to q_1 . S is the new starting state.

A new state (F) is created. q_1 and q_5 have outgoing transition to F. F is the ONLY accepting state.

If we eliminate q_4 , let's consider

Incoming transition from $\rightarrow q_1$

Outgoing transition to $\rightarrow q_3, q_5$

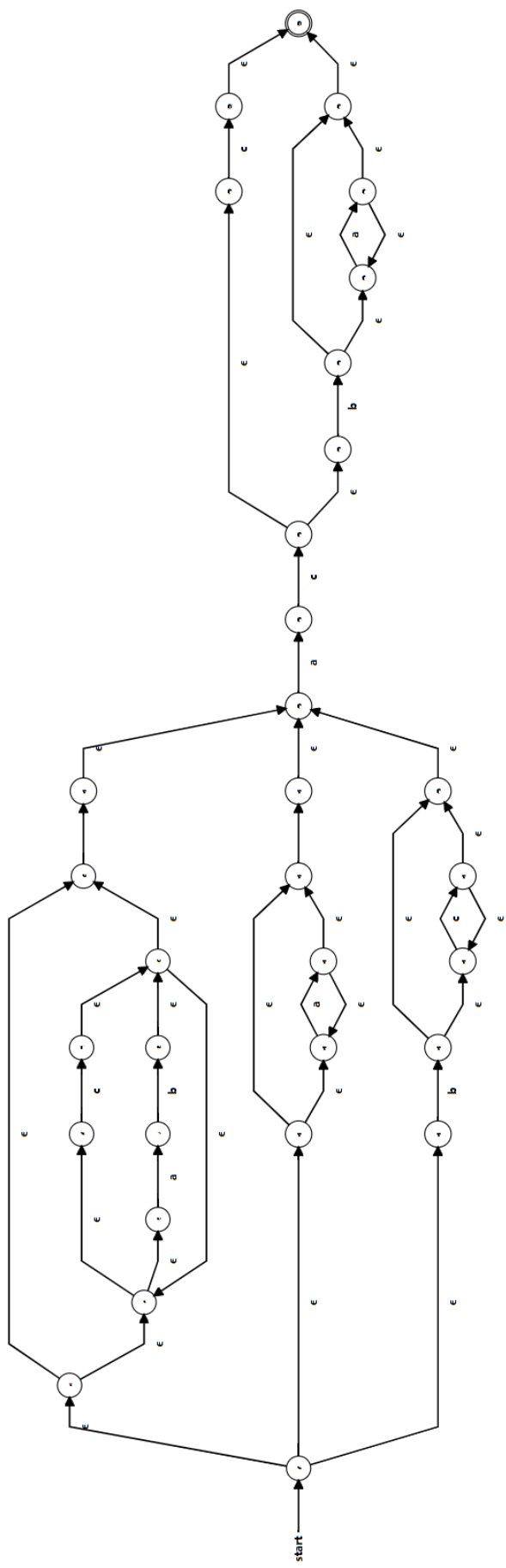
So if q_4 is removed we have, q_1 to q_3 & q_1 to q_5 .

q_1 to $q_3 = aa$

q_1 to $q_5 = ab$

Qs 3) Convert the following Regex to NFA over $\Sigma = \{a,b,c\}$. [8]

$((ab)^+ c)^* | (a^+)^+ bc^+) ac(b.a^+ \cup c)$



Sec11

CSE331 Quiz 03

ID:

SET A

Name:

Sec:

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1x 2 = 2]

a) The pushdown automata uses a queue as a memory unit for operation.

False. Stack is used.

b) The pushdown automata will always cover any regular language. A PDA will accept any string from RL.

True

2] Write context-free grammar for the following languages.

[Total = 8 Marks]

a) $L_1 = \{ w \in \{0,1\}^* : \text{starts with 1 and contains odd 0s} \}$

[2.5]

b) $L_2 = \{ w \in \{a,b\}^* : a^m b^n \text{ where } m \geq 3n + 1 \text{ and } n \geq 0 \}$

[2.5]

c) $L_3 = \{ 0^i 1^j 2^k : j = i + 3k ; i, j, k > 0 \}$

[3]

a) $S \rightarrow 1P$

$P \rightarrow AP0$

$A \rightarrow B0B0B \mid e$

$B \rightarrow 1B \mid e$

b) $S \rightarrow aS \mid aA$

$A \rightarrow aaaAb \mid e$

c)

$A \rightarrow BC$

$B \rightarrow 0B1 \mid 01$

$C \rightarrow 11C2 \mid 1112$

3] Take a look at the grammar below and solve the following problems.

[Total = 10 Marks]

$S \rightarrow AX$

$A \rightarrow 0A \mid 1B \mid \epsilon$

$B \rightarrow 0B \mid 1A$

$X \rightarrow 00X \mid 01X \mid 10X \mid 11X \mid 0 \mid 1$

(a) Give a leftmost derivation for the string 0101110.

[2]

(b) Sketch the parse tree corresponding to the derivation you gave in (a).

[2]

(c) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string 0101110. **[6]**

CSE331 Quiz 03

CSE331 Quiz 03

ID:

SET B

Name:

Sec:

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1x 2 = 2]

a) The pushdown automata uses a stack as a memory unit for operation.

True

b) The pushdown automata will only cover few Regular Language. A PDA will accept few strings of RL, not all.

False. It will accept all RL

2] Write context-free grammar for the following languages.

[Total = 8 Marks]

a) $L_1 = \{ w \in \{c,d\}^* : \text{starts with } c \text{ and contains even } d\}$

[2.5]

b) $L_2 = \{ w \in \{0,1\}^* : 0^p 1^q \text{ where } p \geq 4q + 1 \text{ and } q \geq 0 \}$

[2.5]

c) $L_3 = \{ 0^i 1^j 2^k : j = i + 2k ; i, j, k > 0 \}$

[3]

a) $S \rightarrow cP$

$P \rightarrow AP$

$A \rightarrow BdBdB \mid e$

$B \rightarrow cB \mid e$

b) $S \rightarrow 0S \mid 0A$

$A \rightarrow 0000A1 \mid e$

c)

$A \rightarrow BC$

$B \rightarrow 0B1 \mid 01$

$C \rightarrow 11C2 \mid 112$

3] Take a look at the grammar below and solve the following problems.

[Total = 10 Marks]

$S \rightarrow AX$

$A \rightarrow 0A \mid 1B \mid \epsilon$

$B \rightarrow 0B \mid 1A$

$X \rightarrow 00X \mid 01X \mid 10X \mid 11X \mid 0 \mid 1$

(a) Give a leftmost derivation for the string 1110010.

[2]

(b) Sketch the parse tree corresponding to the derivation you gave in (a).

[2]

(c) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string 1110010. **[6]**

Sec12

ID:

SET A

Name:

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1] Take a look at the grammar below and solve the following problems. **[Total = 10 Marks]**

$$S \rightarrow AX$$

$$A \rightarrow 0A \mid 1B \mid \epsilon$$

$$B \rightarrow 0B \mid 1A$$

$$X \rightarrow 00X \mid 01X \mid 10X \mid 11X \mid 0 \mid 1$$

- (a) Give a leftmost derivation for the string 0101110. **[2]**
- (b) Sketch the parse tree corresponding to the derivation you gave in (a). **[2]**
- (c) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string 0101110. **[6]**

2] Write context-free grammar for the following languages.

[Total = 10 Marks]

a) $L_1 = \{ w \in \{0,1\}^* : \text{starts with 1 and contains odd 0s} \}$

[2]

b) $L_2 = \{ w \in \{a,b\}^* : a^m b^n \text{ where } m \geq 3n + 1 \text{ and } n \geq 0 \}$

[2]

c) $L_3 = \{ 0^i 1^j 2^k : j > i + 3k ; i, j, k > 0 \}$

[3.5]

d) $L_4 = \{ w \in \{0,1\}^* : \text{Even Length Palindrome with the first two characters being 01} \}$ [2.5]

a) $A \rightarrow 1BC$

$B \rightarrow C0C0B \mid 0$

$C \rightarrow 1C \mid \epsilon$

b) $S \rightarrow aS \mid aA$

$A \rightarrow aaaAb \mid \epsilon$

c)

$A \rightarrow BDC$

$B \rightarrow 0B1 \mid 01$

$C \rightarrow 111C2 \mid 1112$

$D \rightarrow 1D \mid 1$

* Without D, $j = i + 3k$ is reached. Using D, we have atleast extra 1, so $j > i + 3k$ is reached.

d)

$A \rightarrow 01B10$

$B \rightarrow 1B1 \mid 0A0 \mid \epsilon$

ID:

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1] Take a look at the grammar below and solve the following problems. **[Total = 10 Marks]**

$$S \rightarrow AX$$

$$A \rightarrow 0A \mid 1B \mid \epsilon$$

$$B \rightarrow 0B \mid 1A$$

$$X \rightarrow 00X \mid 01X \mid 10X \mid 11X \mid 0 \mid 1$$

(a) Give a leftmost derivation for the string 1110010.

[2]

(b) Sketch the parse tree corresponding to the derivation you gave in (a).

[2]

(c) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string 1110010. **[6]**

2] Write context-free grammar for the following languages.

[Total = 10 Marks]

a) $L_1 = \{ w \in \{0,1\}^* : \text{starts with 0 and contains even 0s} \}$

[2]

b) $L_2 = \{ w \in \{a,b\}^* : a^p b^q \text{ where } p \geq 4q + 1 \text{ and } q \geq 0 \}$

[2]

c) $L_3 = \{ 0^i 1^j 2^k : j > i + 2k ; i, j, k > 0 \}$

[3.5]

d) $L_4 = \{ w \in \{0,1\}^* : \text{Odd Length Palindrome with the first two characters being 11} \}$ [2.5]

a)

$A \rightarrow 0BC$

$B \rightarrow C0C0B$

$C \rightarrow 1C \mid \epsilon$

b) $S \rightarrow 0S \mid 0A$

$A \rightarrow 0000A1 \mid \epsilon$

c)

$A \rightarrow BDC$

$B \rightarrow 0B1 \mid 01$

$C \rightarrow 11C2 \mid 112$

$D \rightarrow 1D \mid 1$

* Without D, $j = i + 3k$ is reached. Using D, we have atleast extra 1, so $j > i + 3k$ is reached.

d) $A \rightarrow 11C11 \mid 111$

$B \rightarrow 1B1 \mid 0B0 \mid 0 \mid 1$

Slot 1

ID:
Name:

SET A

Sec:

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1 x 3 = 3]

a) For decidable language, the Turing Machine halts on every input.

True

b) Pushdown automata only recognize non-regular languages

False. It can do regular language as well.

c) Regex can produce non regular language

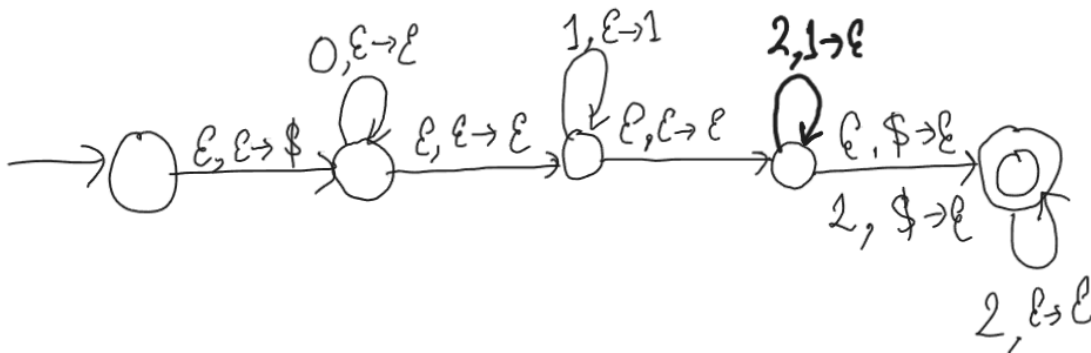
False. It can only produce regular language

2] Draw the PDA for the following language.

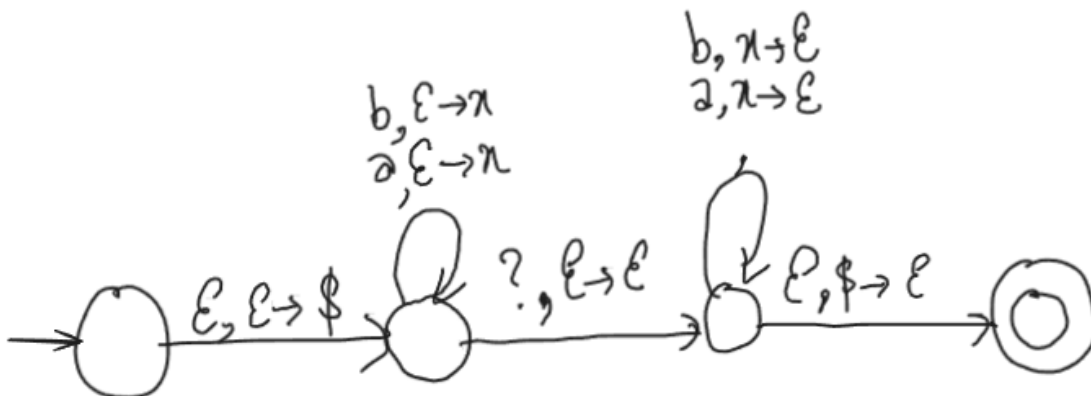
$L_1 = \{ w \in \{0, 1, 2\}^* \mid 0^i 1^j 2^k \text{ where } j \leq k \text{ \& } i, j, k \geq 0 \}$ [4]

$L_2 = \{ w_1, w_2 \in \{a, b\}^* \mid w_1 w_2 \text{ where } |w_1| = |w_2| \}$ [3]

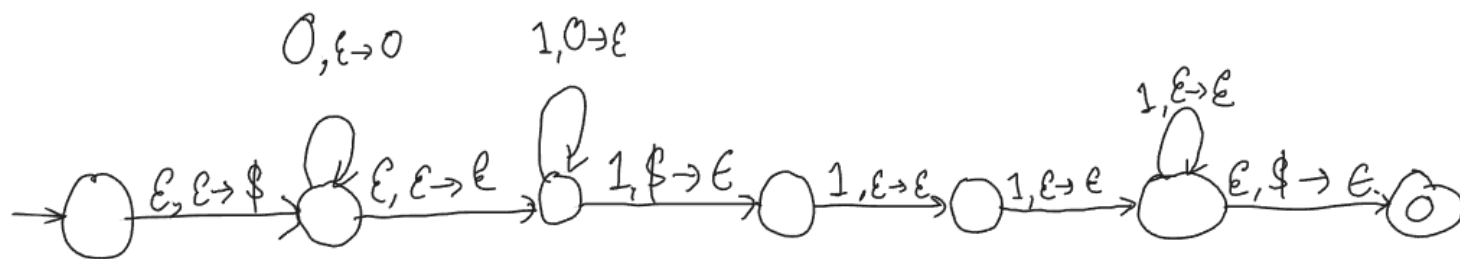
$L_3 = \{ w \in \{0, 1\}^* \mid 0^m 1^n \text{ where } n > m + 2 \}$ [4]



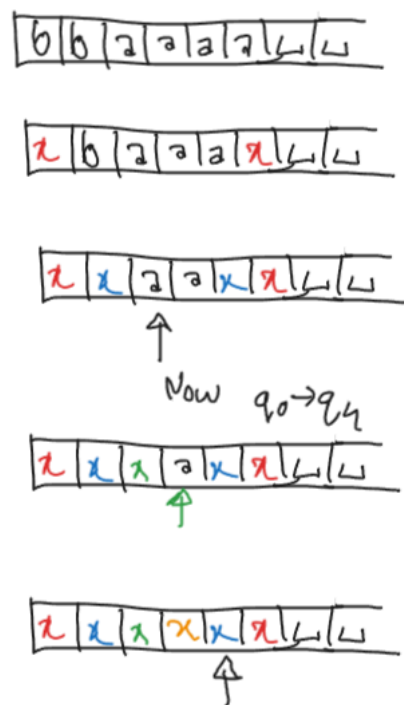
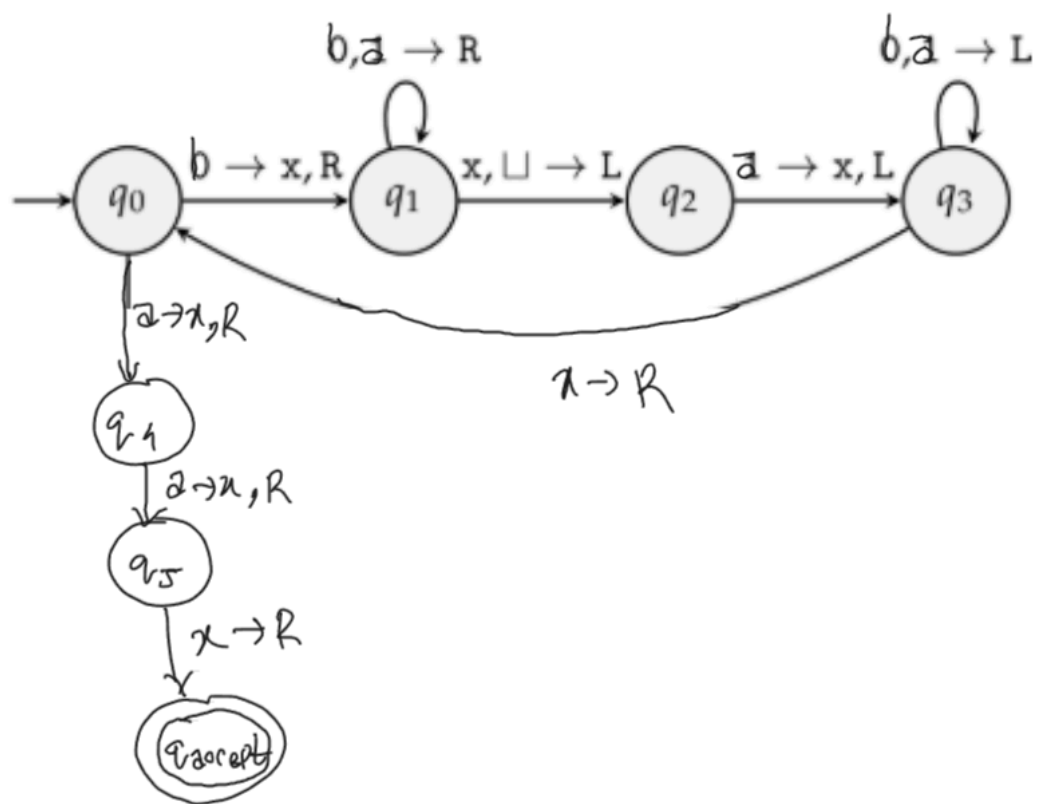
L_2



v



3] Draw a Turning Machine for the language $b^n a^{n+2}$, where $n \geq 0$. [6]



ID:

SET B

Name:

Sec:

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1 x 3 = 3]

a) For recognizable language, the Turing Machine halts on every input.

False . It can halt also.

b) Pushdown Automata recognize every regular and non-regular language

True

c) Context-free grammar cannot produce regular language.

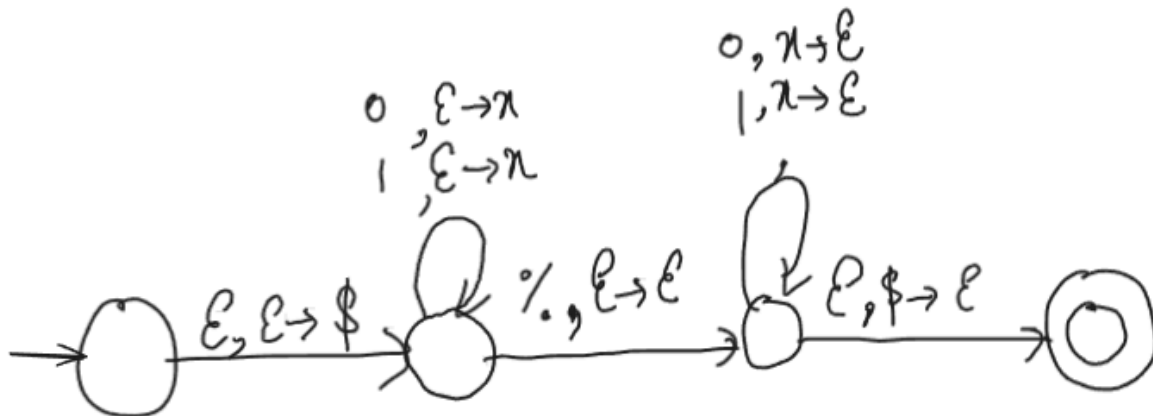
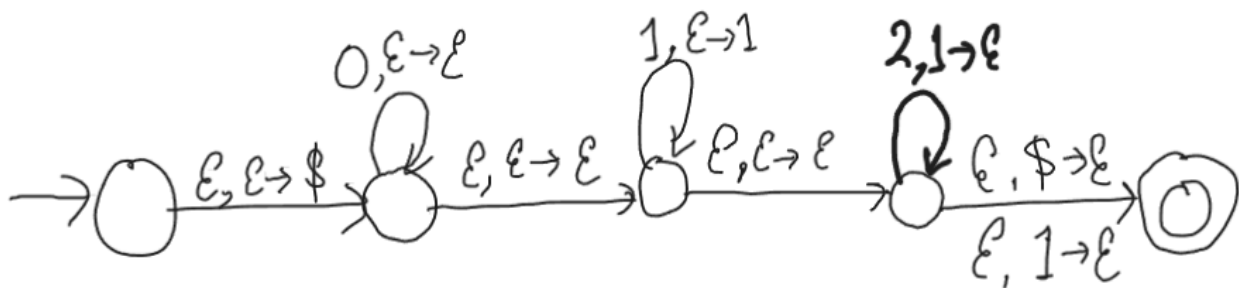
False. It can produce regular language.

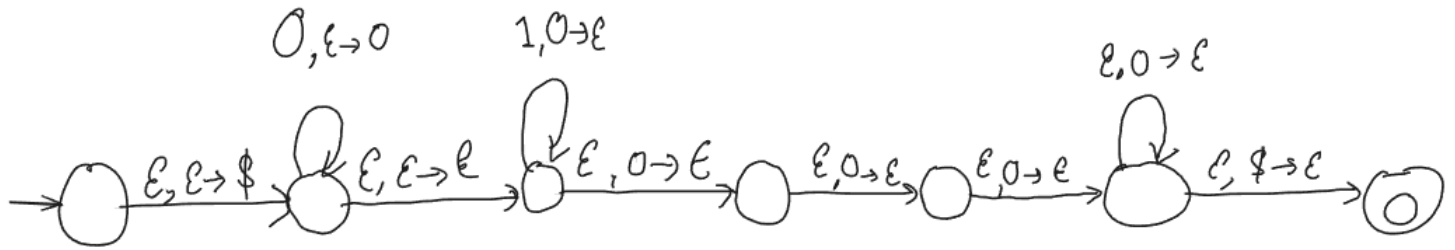
2] Draw the PDA for the following language.

$L_1 = \{ w \in \{0, 1, 2\} \mid 0^i 1^j 2^k \text{ where } j \geq k \text{ \& } i, j, k \geq 0 \}$ [4]

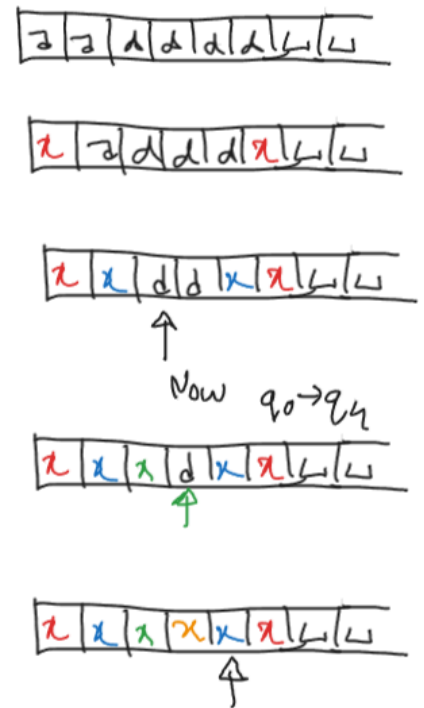
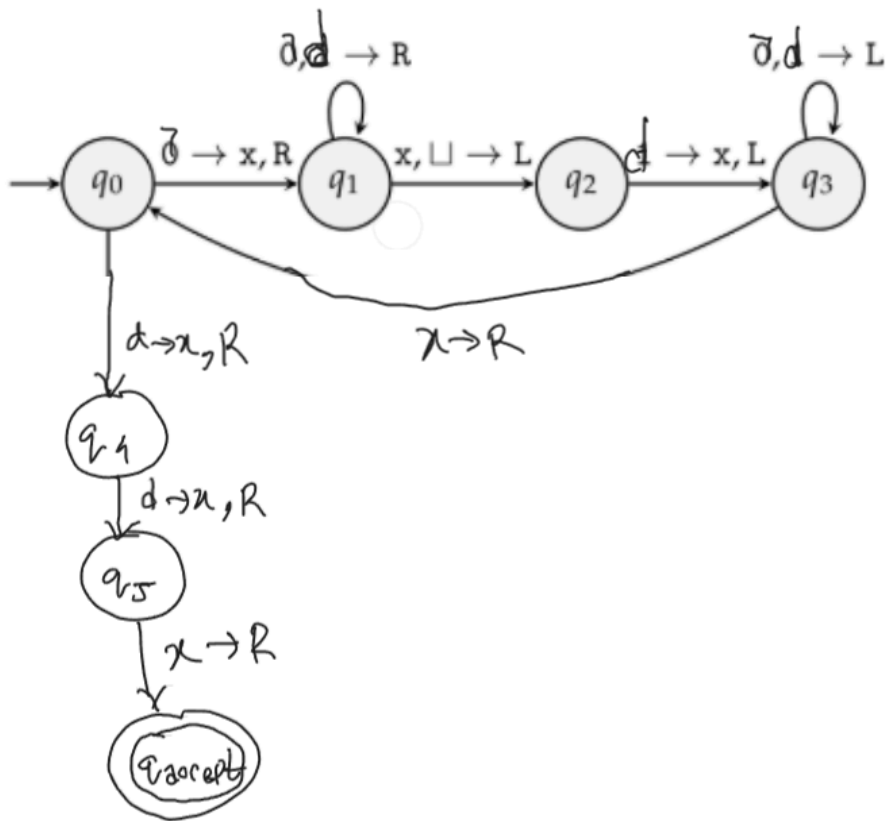
$L_2 = \{ w_1, w_2 \in \{0, 1\} \mid w_1 \% w_2 \text{ where } |w_1| = |w_2| \}$ [3.5]

$L_3 = \{ w \in \{0, 1\} \mid 0^m 1^n \text{ where } m > n + 2 \}$ [3.5]

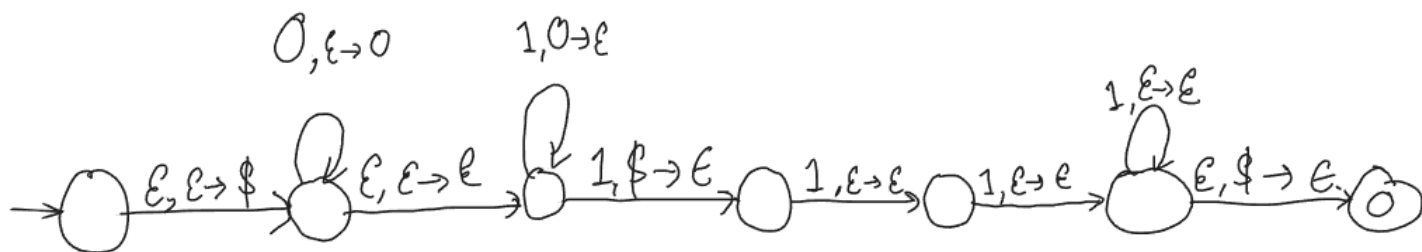




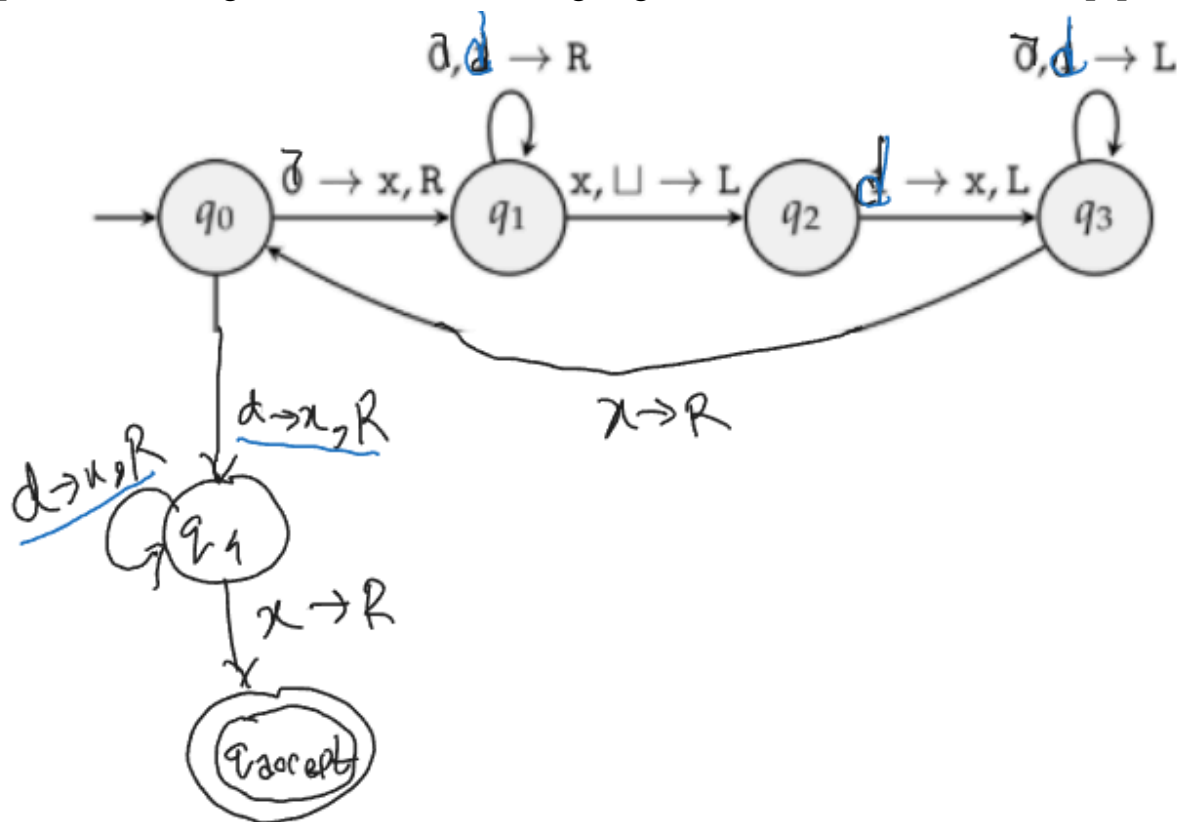
3] Draw a Turning Machine for the language $a^n d^{n+2}$, where $n \geq 0$. [6]



Slot 2



3] Draw a turing machine for the language $a^m d^n$, $n > m$ where $n > 0$ [6]



ID:

SET B

Name:

Sec:

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1 x 3 = 3]

a) For decidable language, the Turing Machine does not halt for all inputs.

False. It halts on all inputs.

b) Pushdown Automata recognize every regular and non-regular language

True

c) Regex can produce non regular language

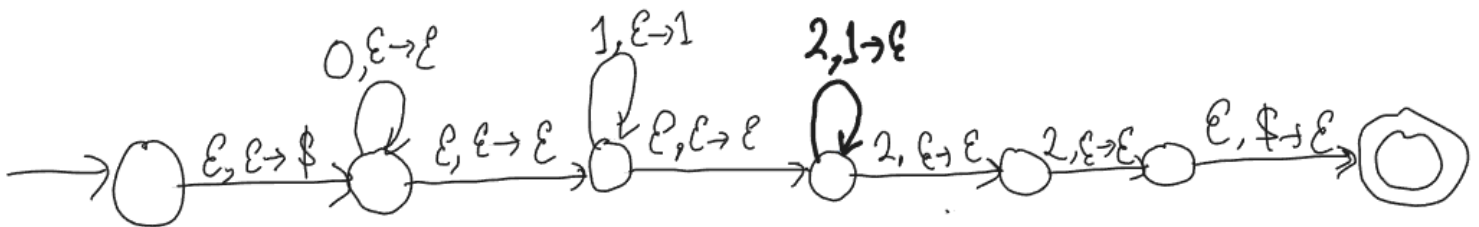
False. It only creates regular language

2] Draw the PDA for the following language.

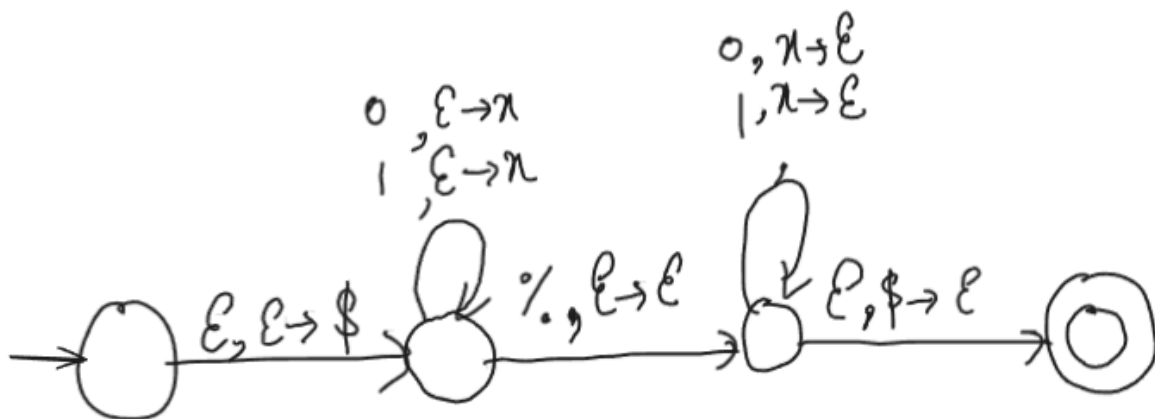
$L_1 = \{ w \in \{0, 1, 2\} \mid 0^i 1^j 2^k \text{ where } k=j+2 \text{ \& } i,j,k \geq 0 \}$ [4]

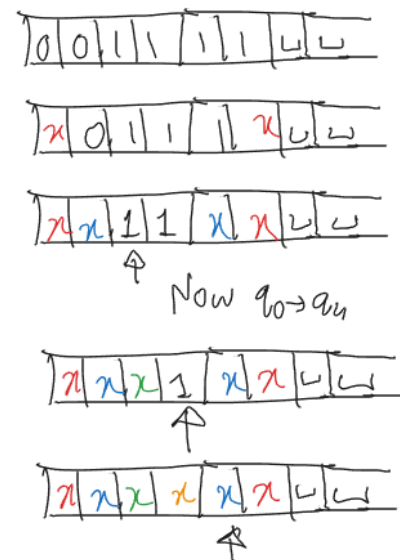
$L_2 = \{ w_1, w_2 \in \{0, 1\} \mid w_1 \% w_2 \text{ where } |w_1| = |w_2| \}$ [3.5]

$L_3 = \{ w \in \{0, 1\} \mid 0^m 1^n \text{ where } m > n+2 \}$ [3.5]



L2:





Sec 11

CSE331 Quiz 05

ID:
Name:

Set A

Sec:

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1 x 3 = 3]

a) Reducibility is converting a simple problem to a complex one to solve and get the solution.

False. It is converting complex problem to simpler ones.

b) All recognizable languages are decidable.

False. All decidable problems are recognizable

c) Context-free grammar will never have a terminal on the left side of the arrow.

True

2] Take a look at the grammar below and solve the following problems. [Total = 6 Marks]

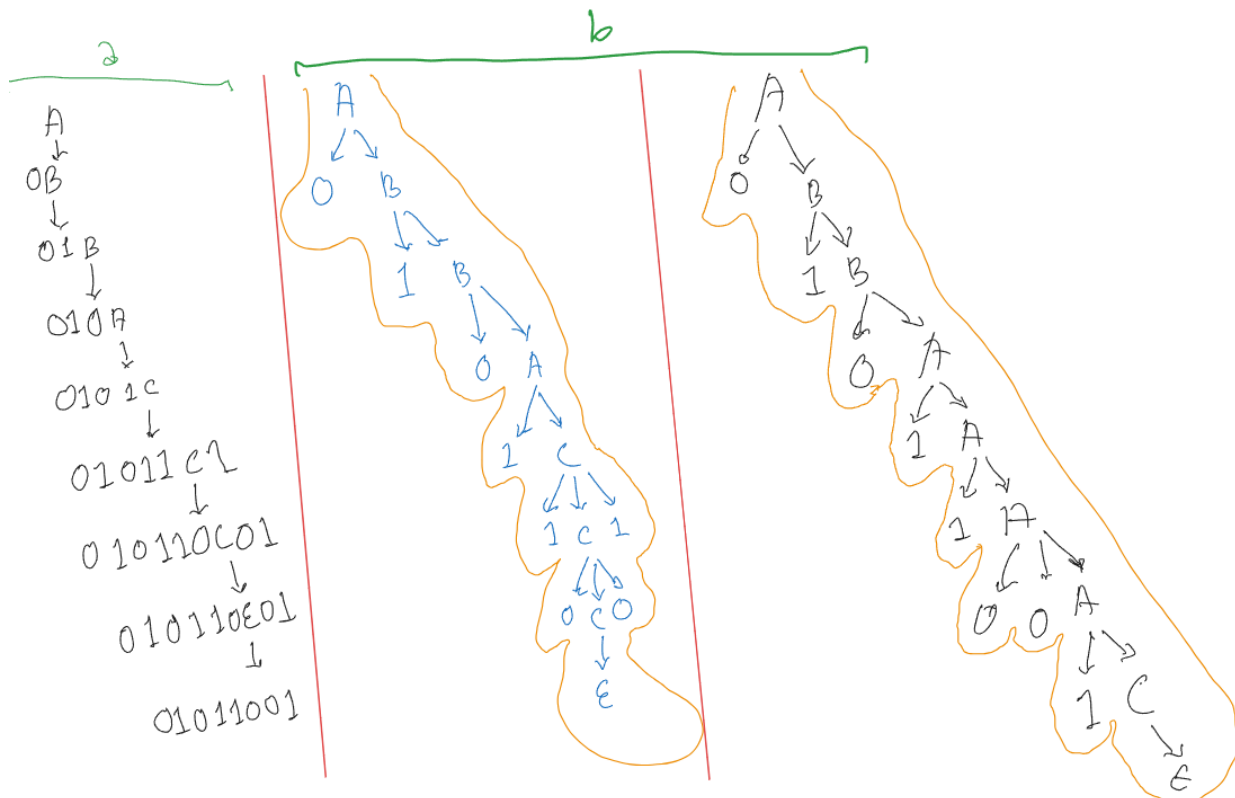
$$A \rightarrow 1A \mid 1C \mid 0B \mid 00A$$

$$B \rightarrow 0A \mid 1B \mid 00B$$

$$C \rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \epsilon$$

(a) Give a leftmost derivation for the string 01011001. [2]

(b) Demonstrate that the given grammar is ambiguous by showing two parse trees for the same string 01011001. [4]



CSE331 Quiz 05

3] Write CFG for the following languages: [2.5 * 2 = 5]

a) $L = \{w \in \{0,1\}^* : \text{Every 0 is followed by odd 1s}\}$

b) $L = \{w \in \{0,1,2\}^* : 0^i 1^j 2^k \text{ where } j \geq k \text{ \& } i, j, k \geq 0\}$

a) $A \rightarrow 1A \mid B \mid \epsilon$

$B \rightarrow 0C$

$C \rightarrow 11C \mid 1 \mid 1A$

b)

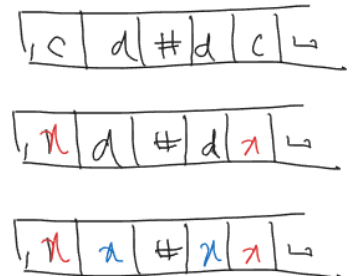
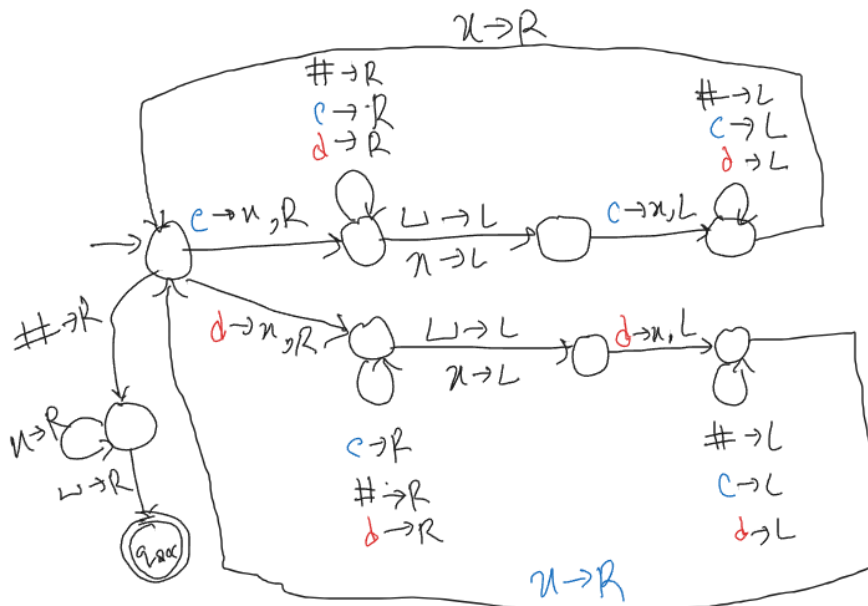
$S \rightarrow CA$

$C \rightarrow 0C \mid \epsilon$

$A \rightarrow 1A \mid 1B \mid B$

$B \rightarrow 1B2 \mid \epsilon$

3] Draw the Turing Machine for the language $\{w \in \{c,d\}^* : w\#w^R\}$: [6]



CSE331 Quiz 05

ID:

Set B

Name:

Sec:

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1 x 3 = 3]

a) Reducibility is converting a simple problem to a complex one to solve and get the solution.

True

b) All decidable languages are recognizable.

True

c) Context-free grammar will never always have a variable on the right side of arrow.

False

2] Take a look at the grammar below and solve the following problems. **[Total = 6 Marks]**

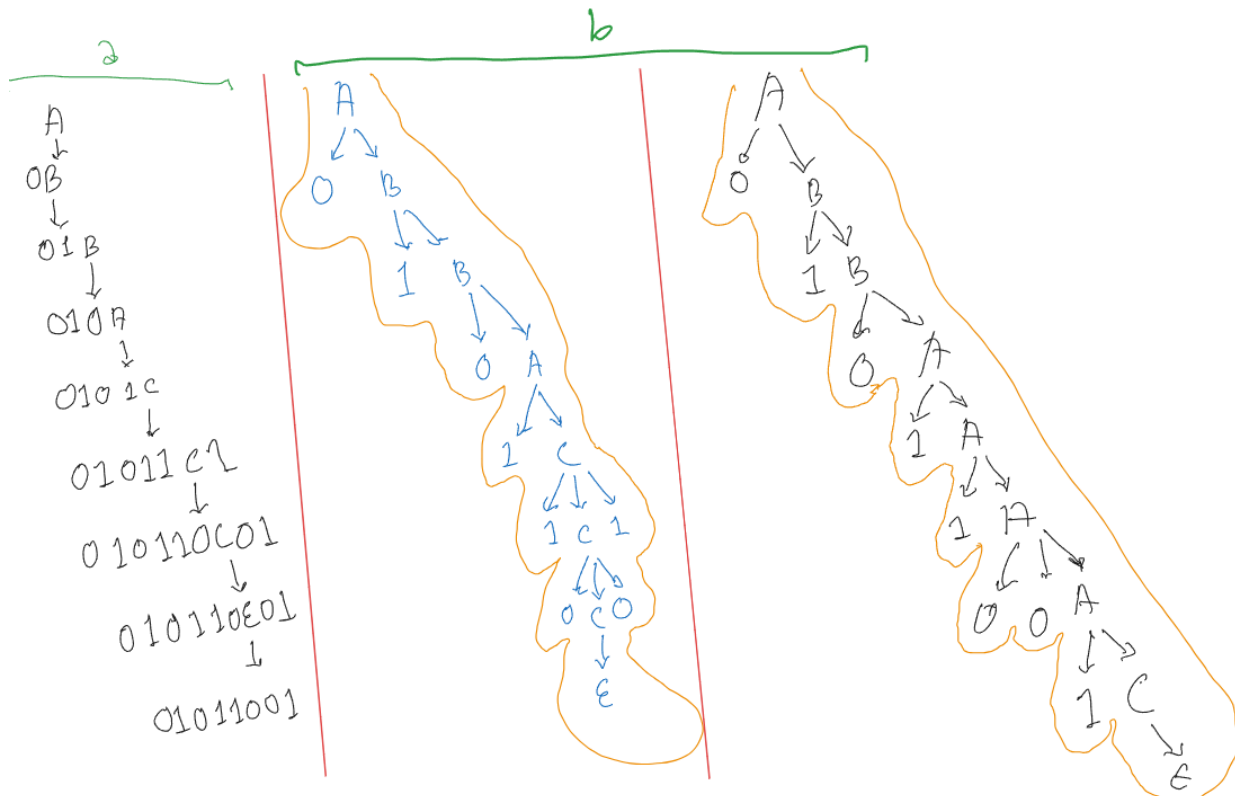
$$A \rightarrow 1A \mid 1C \mid 0B \mid 00A$$

$$B \rightarrow 0A \mid 1B \mid 00B$$

$$C \rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \epsilon$$

(a) Give a leftmost derivation for the string 01011001. **[2]**

(b) Demonstrate that the given grammar is ambiguous by showing two parse trees for the same string 01011001. **[4]**



CSE331 Quiz 05

3] Write CFG for the following languages: [2.5 * 2 = 5]

a) $L = \{w \in \{0,1\}^* : \text{Every 1 is followed by odd 0s}\}$

b) $L = \{w \in \{0,1,2\}^* : 0^i 1^j 2^k \text{ where } j \leq k \text{ \& } i,j,k \geq 0\}$

a) $A \rightarrow 0A \mid B \mid \epsilon$

$B \rightarrow 1C$

$C \rightarrow 00C \mid 0 \mid 0A$

b) $S \rightarrow CA$

$C \rightarrow 0C \mid \epsilon$

$A \rightarrow A2 \mid B2 \mid B$

$B \rightarrow 1B2 \mid \epsilon$

3] Draw the Turing Machine for the language $\{w \in \{d,e\}^* : w \# w^R\}$: [6]

Similar to Set A , qs 3

Sec12

3] Write CFG for the following languages: [2 * 2 = 4]

a) $L = \{w \in \{0,1\}^* : \text{Every 0 is followed by odd 1s}\}$

b) $L = \{w \in \{0,1,2\}^* \mid 0^i 1^j 2^k \text{ where } j=k+i+3 \text{ \& } i,j,k \geq 0\}$

a) $A \rightarrow 1A \mid B \mid e$

$B \rightarrow 0C$

$C \rightarrow 11C \mid 1 \mid 1A$

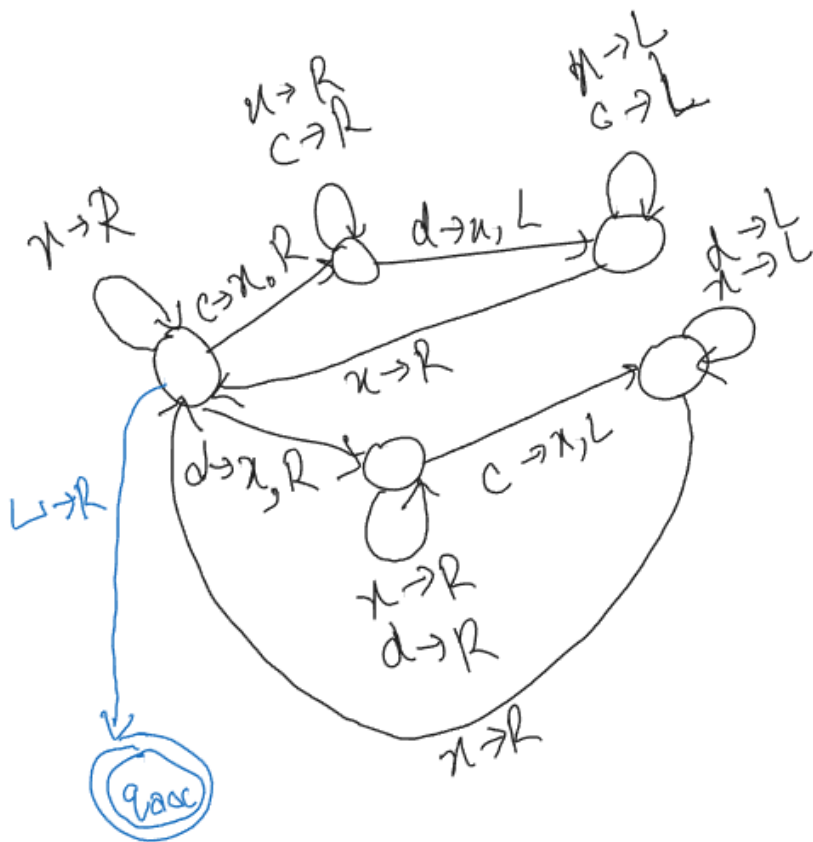
b)

$A \rightarrow B111C$

$B \rightarrow 0B1 \mid e$

$C \rightarrow 1C2 \mid e$

3] Draw the Turing Machine for the language $\{w \in \{c,d\}^* : \text{equal number of 'c' and 'd'}\}$: [6]



c	d	d	c	␣
---	---	---	---	---

x	x	d	c	␣
---	---	---	---	---

x	x	x	x	␣
---	---	---	---	---

ID:

SET B

Name:

Sec:

1] Write True / False. If it is **False**, you MUST write the correct answer with a one-line explanation. [1 x 3 = 3]

a) Tape alphabet will never be same as the alphabet for a language in Turing Machine.

True

b) All decidable languages are recognizable.

True

c) A M_{DFA} is undecidable. Do recall, M here is a Turing machine.

False. It is decidable

2] Take a look at the grammar below and solve the following problems.

[Total = 6 Marks]

$$A \rightarrow 1A \mid 1C \mid 0B \mid 00A$$

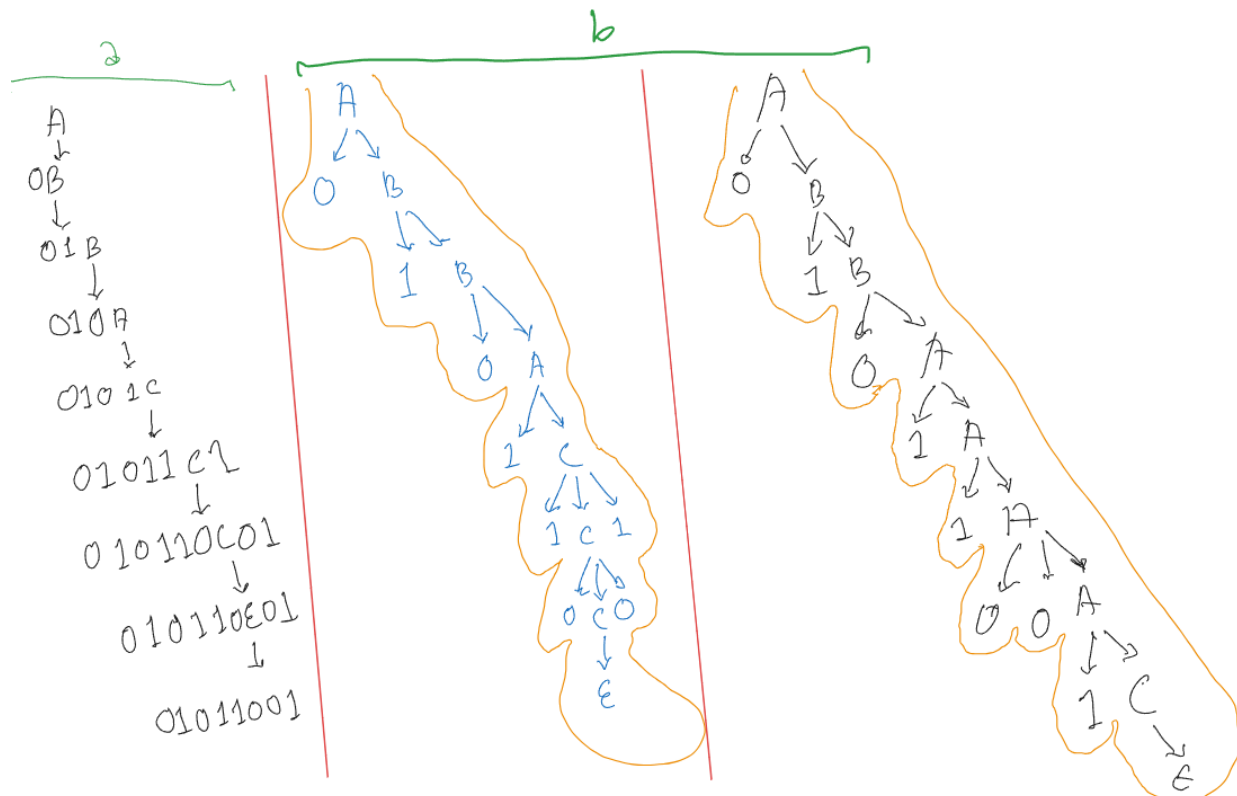
$$B \rightarrow 0A \mid 1B \mid 00B$$

$$C \rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \epsilon$$

(a) Give a leftmost derivation for the string 01011001.

[2]

(b) Demonstrate that the given grammar is ambiguous by showing two parse trees for the same string 01011001. **[4]**



3] Write CFG for the following languages: [2 * 2 = 4]

a) $L = \{w \in \{0,1\}^* : \text{Every 1 is followed by odd 0s}\}$

b) $L = \{w \in \{0, 1, 2\}^* \mid 0^i 1^j 2^k \text{ where } j=k+i+2 \text{ \& } i,j,k \geq 0\}$

a) $A \rightarrow 0A \mid B \mid e$

$B \rightarrow 1C$

$C \rightarrow 00C \mid 0 \mid 0A$

b)

$A \rightarrow B11C$

$B \rightarrow 0B1 \mid e$

$C \rightarrow 1C2 \mid e$

3] Draw the Turing Machine for the language $\{w \in \{d, f\}^* : \text{equal number "d" and "f"}\}$: [7]

Similar to Set A , qs 3